

合成関数の微分

$y=f(u)$, $u=g(x)$ が微分可能であるとき、合成関数 $y=f(g(x))$ も微分可能で

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \leftarrow \text{「x-u」は割り}$$

また, $\frac{dy}{du} = f'(u)$, $\frac{du}{dx} = g'(x)$ であるから

$$\frac{dy}{dx} f(g(x)) = f'(g(x)) g'(x) \quad \leftarrow \text{「f(□)」' = f'(□) \cdot □'}$$

外微分 中微分

(例) 次の関数を微分せよ。

(1) $y = (x^2+1)^2 \quad \leftarrow y = \square^2 \text{ 扱い} \quad y' = 2\square \cdot \square'$

$$y' = 2(x^2+1) \cdot (x^2+1)'$$

$$= 4x(x^2+1) //$$

(2) $y = \sqrt{1-x^2} \quad \leftarrow y = \square^{\frac{1}{2}} \text{ 扱い} \quad y' = \frac{1}{2} \cdot \square^{-\frac{1}{2}} \cdot \square' = \frac{\square'}{2\square}$

$$y' = \frac{(1-x^2)'}{2\sqrt{1-x^2}}$$

$$= -\frac{x}{\sqrt{1-x^2}} //$$

(3) $y = \frac{1}{(x^3+1)^2} \quad \leftarrow y = \square^{-2} \text{ 扱い} \quad y' = -2\square^{-3} \cdot \square'$

$$y' = -2(x^3+1)^{-3} \cdot (x^3+1)'$$

$$= -\frac{6x^2}{(x^3+1)^3} //$$

(4) $y = \frac{1}{3\sqrt{x^2+1}} \quad \leftarrow y = \square^{-\frac{1}{3}} \text{ 扱い} \quad y' = -\frac{1}{3} \cdot \square^{-\frac{4}{3}} \cdot \square'$

$$y' = -\frac{1}{3} (x^2+1)^{-\frac{4}{3}} (x^2+1)'$$

$$= -\frac{2x}{3^{\frac{3}{2}}(x^2+1)^{\frac{4}{3}}} //$$

(5) $y = \sin\left(2x + \frac{\pi}{6}\right) \quad \leftarrow y = \sin \square \text{ 扱い} \quad y' = \cos \square \cdot \square'$

$$y' = \cos\left(2x + \frac{\pi}{6}\right) (2x + \frac{\pi}{6})'$$

$$= 2\cos\left(2x + \frac{\pi}{6}\right) //$$

(6) $y = \cos^3 x \quad \leftarrow y = \square^3 \text{ 扱い} \quad y' = 3\square^2 \cdot \square'$

$$y' = 3\cos^2 x \cdot (\cos x)'$$

$$= -3\cos^2 x \sin x //$$

(7) $y = \log_2 2x \quad \leftarrow y = \log \square \text{ 扱い} \quad y' = \frac{1}{\square} \cdot \square'$

$$y' = \frac{1}{2x} (2x)'$$

$$= \frac{1}{x} //$$

(8) $y = \log_2 (x^2+1) \quad \leftarrow y = \log_2 \square \text{ 扱い} \quad y' = \frac{1}{\square \log 2} \cdot \square'$

$$y' = \frac{1}{(x^2+1) \log 2} \cdot (x^2+1)'$$

$$= \frac{2x}{(x^2+1) \log 2} //$$

(9) $y = \log \left| \frac{2x-1}{2x+1} \right|$

$$y = \log |2x-1| - \log |2x+1| \quad \leftarrow y = \log \square - \log \square \text{ 扱い} \quad y' = \frac{1}{\square} \square' - \frac{1}{\square} \square'$$

扱い

$$y' = \frac{1}{2x-1} (2x-1)' - \frac{1}{2x+1} (2x+1)'$$

$$= \frac{2}{2x-1} - \frac{2}{2x+1}$$

$$= \frac{4}{4x^2-1} //$$

(10) $y = e^{x^2} \quad \leftarrow y = e^{\square} \text{ 扱い} \quad y' = e^{\square} \cdot \square'$

$$y' = e^{x^2} (x^2)'$$

$$= 2xe^{x^2} //$$